MD RAKIBUL ISLAM RAIHAN

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LinkedIn: https://www.linkedin.com/in/theraihanrakib
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Seeking Positions: AI Algorithm R&D Engineer (Multimodal / AIGC / LLM-VLM) | AI Product Manager

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Objective

Master's candidate in Software Engineering (**top 1% of program**) with current industry experience designing **NVFP4 pretraining pipelines for LLMs and VLMs**, optimizing inference serving (**vLLM, TensorRT-LLM, Transformer Engine**), and developing **custom CUDA/Triton kernels** for mixed-precision training.

Research focuses on **multimodal AI**, **generative models (Diffusion, GANs)**, and **audio-visual fusion** — from fundamentals to deployment. Proficient in **PyTorch**, **TensorFlow**, and **Apache Spark** for large-scale data processing. Previously led an **end-to-end text-to-video generation pipeline** in an academic lab and delivered **full-stack AI Agent solutions** with commercial deployment.

Committed to advancing foundation models and multimodal algorithms. Seeking **AI Algorithm R&D Intern** (Multimodal / AIGC / Computer Vision / Robotics) or **AI Product Manager Intern** role bridging cutting-edge research and real-world products to drive impactful innovation.

Education

Master of Engineering in Software Engineering
Northwestern Polytechnical University, Xi'an, China (**QS #499, 985, 211, E9 League**)
Sep 2024 – Mar 2027

- **GPA:** 88/100
- **Rank:** 1%
- **Relevant Coursework:** Machine Learning, Computer Vision, Natural Language Processing, Software Project Management.
- **Extracurricular Activities:** Member of Student Union, Academic Mentor, Teaching Assistant.
- **Research:** "Multimodal Video Generation: Integrating Audio and Visual Signals for Realistic Content Creation" (Under the guidance of Dr. Zhang Peng).
- **Thesis:** "Detecting Deepfake Videos: A Multimodal AI Framework Leveraging Visual, Auditory, and Temporal Inconsistencies." (Under the guidance of Dr. Zhang Peng).

Bachelor of Engineering in Computer Science and Technology
Northwestern Polytechnical University, Xi'an, China (**QS #499, 985, 211, E9 League**)
Sep 2020 – Jul 2024

- **GPA:** 85/100
- **Rank:** 1%
- **Outstanding Undergraduate Graduate (Top 10)** (link)
- **Relevant coursework**: Algorithms, Data Structures, Software Engineering, Database Systems.

- **Extracurricular Activities:** Director of International Student Union (2023), Academic Volunteer (2023,2024)
- **Thesis:** "Design and implemented a Distributed Confidential Query Protocol for Apache Spark." (Under the guidance of Dr. Chen Yaxing).

Experience

Graduate Research Assistant
Multimodal Learning and Video Generation Lab, NWPU
Sep 2024 – Present
Advisor: Dr. Zhang Peng

- Spearheaded research in **audio-visual fusion models** to enhance the realism and synchronization of generated video content.
- Designed and implemented a novel **self-supervised learning framework** for cross-modal representation learning, improving alignment between audio and visual data streams.
- Engineered an end-to-end **text-to-video generation pipeline** leveraging Transformer-based architectures (e.g., Diffusion Transformers) to convert semantic prompts into dynamic visual sequences.

Undergraduate Research Intern
High Performance Computing Lab, NWPU
Jan 2023 – Jul 2024
Advisor: Dr. Chen Yaxing

- Researched and implemented a **Distributed Confidential Query Protocol**, designing mechanisms for secure, privacy-preserving data retrieval across distributed nodes.
- Developed large-scale data processing modules for a **cloud computing and networking project** using **Apache Spark and SQL**, optimizing query performance on distributed datasets.

Internship

Technical Staff Intern – AI Infra
InfiX.ai LIMITED
Apr 2026 – Present

- Design and implement NVFP4 pretraining pipelines for LLMs and VLMs, targeting 4-bit low-precision training at scale.
- Optimize inference serving using vLLM, TensorRT-LLM, and Transformer Engine with NVFP4 tensor cores to improve throughput and latency.
- Develop custom CUDA/Triton kernels for packed FP4 data formats and mixed-precision logic, enabling efficient memory access and compute.
- Profile and debug distributed training and inference workloads using Nsight Systems and Nsight Compute to ensure full hardware utilization.
- Validate convergence and model quality of FP4 training against BF16 baselines across multiple architectures and tasks.
- Collaborate on quantization-aware training (QAT) and block-wise scaling strategies to narrow the accuracy gap at extreme low precision.

Software Engineer – AI Agent
Hong Kong Amraim Group Co., Ltd.
Jun 2025 – Sep 2025

- Engineered intelligent AI agents that automated key marketing workflows, reducing manual effort and boosting team productivity by 25%.
- Architected and delivered two end-to-end full-stack applications from concept to deployment, directly addressing core business requirements and user needs.

- Built dynamic, responsive front-end interfaces using React and developed scalable back-end services with RESTful APIs to seamlessly integrate AI/ML features.
- Drove the full software development lifecycle in an Agile (Scrum) environment, from planning to deployment, ensuring the delivery of secure, scalable, and production-ready solutions.

Electrical Software Engineer – Algorithms
Shaanxi Longong Intelligent Technology Co., Ltd.
Feb 2025 – Apr 2025

- Engineered and optimized real-time motor control algorithms for industrial automation machinery, improving system response time by 15% and increasing overall throughput.
- Developed, debugged, and validated embedded control software utilizing PLC programming (Codesys) and industrial communication protocols (Modbus, CAN, Ethernet/IP) to ensure reliable system operation.
- Designed electrical schematics and led prototyping for hardware-software interfaces in new automation equipment, ensuring full compliance with industry safety and regulatory standards (e.g., IEC 61131).
- Collaborated within a cross-functional R&D team using Agile (Scrum) methodologies; managed project tasks in JIRA and maintained version control with Git to streamline the development lifecycle.

Professional & Technical Skills

- **AI/ML & Deep Learning**
  PyTorch • TensorFlow • Keras • Transformer Architectures • Self-Supervised & Multimodal Learning • Audio-Visual Fusion • Generative Models (Diffusion) • NVFP4 Low-Precision Training • vLLM • TensorRT-LLM • Transformer Engine • Quantization-Aware Training (QAT) • Block-Wise Scaling
- **Systems & High-Performance Computing**
  CUDA • Triton (Kernel Development) • NVIDIA Nsight Systems • NVIDIA Nsight Compute • Mixed-Precision Training • Distributed Model Fusion • FP8/FP4 Optimization
- **Programming & Data Engineering**
  Python (Proficient) • C++ • Java • SQL • MATLAB • Apache Spark • Hadoop • Hive • RESTful APIs
- **Cloud, MLOps & Embedded Systems**
  AWS • Google Cloud Platform (GCP) • Docker • Git • Embedded Systems (PLC, Codesys, Modbus/CAN/Ethernet/IP)
- **Methodologies & Core Competencies**
  Agile/Scrum Project Management • Cross-Functional Collaboration • Full Software/Research Lifecycle • Technical Research & Writing • Complex Problem-Solving • Electrical Prototyping & Safety Compliance • Performance Profiling & Debugging

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Projects

1. Alcohol Detection System for Automobiles | *Embedded IoT Project*

- Engineered a real-time alcohol detection prototype for automotive safety using an IR sensor and an Arduino/ESP32 microcontroller.
- Developed the sensor data processing firmware and a threshold-based alert system to trigger warnings upon detection.
- Designed the hardware interface and enclosure, ensuring reliability for in-vehicle environmental conditions.

2. Distributed Confidential Query Protocol for Apache Spark | *Privacy-Preserving Big Data*

- Designed and implemented a secure, privacy-preserving query execution engine for Apache Spark using homomorphic encryption.
- Maintained 95% data confidentiality with minimal performance overhead during testing on a 10-node cluster.
- Optimized distributed query performance by 20% compared to baseline privacy frameworks.

3. Multimodal Text-to-Video Generation Tool | *Generative AI Research*

- Built an end-to-end text-to-video generation pipeline using Diffusion Transformer (DiT) architectures and GAN components.
- Integrated cross-modal (audio-visual) alignment techniques, with generated outputs achieving 90% user satisfaction in qualitative evaluations.
- Deployed the scalable inference service on AWS EC2 & S3, enabling on-demand video synthesis.

4. GAN-Based Medical Image Enhancement | *Healthcare AI Application*

- Developed a conditional GAN model to super-resolve and denoise low-resolution MRI scans, enhancing clinical detail.
- The enhanced images improved downstream tumor detection accuracy to 85% when validated with a standard segmentation model.

5. Real-Time Visual Object Tracking for UAVs | *Computer Vision & Autonomous Systems*

- Designed and trained a deep learning-based visual tracking system for unmanned aerial vehicles (UAVs) in dynamic environments.
- Achieved 95% tracking accuracy (IoU) when evaluated on the BAIR Robot Pushing Dataset, demonstrating robustness to occlusion.

Certifications

- **AWS Certified Machine Learning – Specialty | Amazon Web Services**
  (Validates ability to ship production ML solutions on a major cloud platform)
- **Deep Learning Specialization | DeepLearning.AI (Coursera)**
  (Certifies core AI/ML knowledge for informed technical decision-making and roadmap planning)
- **Agile & Product Methodology**
  Professional Scrum Master I (PSM I) | Scrum.org
  (*Demonstrates formal Agile/Scrum expertise for leading technical teams and managing development cycles)*

Awards and Achievements

Scholarships & Academic Honors

- Chinese Government Scholarship (2024 – 2026 | Awarded for academic excellence)
- NWPU Presidential Scholarship (2020 – 2023 | Awarded consecutively for top-tier academic performance)
- Sanqin Scholarship (2023 | Merit-based award for outstanding research potential)
- Wu Yajun Scholarship (2024 | Merit-based award for outstanding research potential)

Leadership & Service

- Director, International Students Union (2023 | Led initiatives and represented a diverse student body)
- Academic Volunteer (2022 – 2023 | Assisted in tutoring and research support for underclassmen)

Competitions & Recognition

- Hackathon Winner, NWPU AI Challenge (2022 | Led team to victory in 48-hour applied AI competition)
- Best Undergraduate Thesis Award – *Distributed Confidential Query Protocol for Spark* (Awarded for top research contribution and implementation)

杨瀚

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国籍：孟加拉国
语言：英语（流利，雅思 8.0），中文（日常交流，HSK 3 级）
领英：https://www.linkedin.com/in/theraihanrakib
作品集：https://theraihanrakibb.github.io/Online-Portfolio/
求职意向：AI 算法研发工程师（多模态 / AIGC / 大语言模型-视觉语言模型）| AI 产品经理

目标

软件工程硕士研究生（**专业排名前 1%**），具备当前行业经验，设计 **NVFP4 预训练流水线**用于**大语言模型**和**视觉语言模型**，优化推理服务（**vLLM、TensorRT-LLM、Transformer Engine**），并为**混合精度训练**开发定制 **CUDA/Triton 内核**。

研究方向聚焦**多模态 AI**、**生成模型**（**扩散模型、生成对抗网络**）以及**音视频融合**——从基础理论到实际部署。熟练掌握 **PyTorch**、**TensorFlow** 及大规模数据处理工具 **Apache Spark**。曾在学术实验室主导端到端**文本生成视频**流水线的设计，并交付了具备商业部署能力的全栈 **AI 智能体**解决方案。

致力于推进**基础模型**与**多模态算法**的发展。寻求 **AI 算法研发实习生**（**多模态 / AIGC / 计算机视觉 / 机器人**）或 **AI 产品经理实习生**岗位，结合前沿研究与实际产品，推动有影响力的创新。

教育背景

软件工程硕士
西北工业大学，西安 中国 **(QS 全球排名第 499 位 · 985 工程 · 211 工程 · 卓越联盟成员)**
2024 年 9 月 – 2027 年 3 月

- **GPA：** 88/100
- **排名：** 1%
- **相关课程：** 机器学习、计算机视觉、自然语言处理、软件项目管理。
- **课外活动：**学生会成员、学业导师、助教。
- **研究：**《多模态视频生成：整合音频与视觉信号以实现真实内容创作》（在张蓬博士指导下）。
- **论文：**“ 检测深度伪造视频：利用视觉、听觉和时间不一致的多模态人工智能框架。”（在张蓬博士指导下）

计算机科学与技术工程学士
西北工业大学，西安 中国 **(QS 全球排名第 499 位 · 985 工程 · 211 工程 · 卓越联盟成员)**
2020 年 9 月 – 2024 年 7 月

- **GPA：** 85/100
- **排名：** 1%

- **优秀本科毕业生（前十名）（链接）**
- **相关课程**：算法、数据结构、软件工程、数据库系统。
- **课外活动：**国际学生会主任（2023 年）、学术志愿者（2023 年、2024 年）
- **论文：**“设计并实现了 Apache Spark 的分布式机密查询协议。”（在陈亚兴博士指导下）

经历

研究生助理
多模态学习与视频生成实验室，西北工业大学
2024 年 9 月 – 至今
导师：张蓬博士

- 主导了视**听融合模型**的研究，以提升生成视频内容的真实性和同步性。
- 设计并实施了一种创新**的自监督学习框架**，用于跨模态表示学习，改善了音频和视觉数据流之间的对齐。
- 利用基于 Transformer 的架构（如 Diffusion Transformers）**设计了端到端**文本到视频生成流程，将语义提示转换为动态视觉序列。

本科生研究实习生
西北工业大学高性能计算实验室
2023 年 1 月 – 2024 年 7 月
导师：陈雅兴博士

- 研究并实施了**分布式机密查询协议**，设计了跨分布节点安全、保护隐私的数据检索机制。
- 为云计算和网络项目**开发了大规模数据处理模块，**利用 **Apache Spark 和 SQL** 优化分布式数据集的查询性能。

实习经历

技术员工实习生 – AI 基础设施
InfiX.ai LIMITED (无界智索(深圳)科技有限责任公司)
2026 年 4 月 – 至今

- 设计并实现用于大语言模型和视觉语言模型的 NVFP4 预训练流水线，面向规模化 4 位低精度训练
- 使用 vLLM、TensorRT-LLM 和 Transformer Engine 优化推理服务，利用 NVFP4 张量核心提升吞吐量并降低延迟
- 开发针对压缩 FP4 数据格式和混合精度逻辑的定制 CUDA/Triton 内核，实现高效内存访问与计算
- 使用 Nsight Systems 和 Nsight Compute 对分布式训练与推理负载进行性能分析与调试，确保硬件充分利用
- 在多种架构与任务上验证 FP4 训练相较于 BF16 基线的收敛效果与模型质量
- 合作研究量化感知训练（QAT） 和分块缩放策略，缩小极低精度下的精度差距

软件工程师 – AI 智能体
Hong Kong Amraim Group Co., Ltd. (香港安莱姆集团有限公司)
2025 年 6 月 – 2025 年 9 月

- 设计了智能人工智能代理，自动化关键营销工作流程，减少人工工作，提升团队生产力 25%。
- 从构想到部署，构建并交付了两个端到端的全栈应用，直接满足核心业务需求和用户需求。

- 利用 React 构建了动态响应式的前端接口，并开发了可扩展的后端服务，采用 RESTful API，实现了 AI/ML 功能的无缝集成。
- 在敏捷（Scrum）环境中推动整个软件开发生命周期，从规划到部署，确保安全、可扩展且具备生产环境的解决方案交付。

电气软件工程师
Shaanxi Longong Intelligent Technology Co., Ltd. (陕西龙宫智能科技有限公司)
2025 年 2 月– 2025 年 4 月

- 为工业自动化机械设计并优化了实时电机控制算法，提高了系统响应时间 15%，提高了整体吞吐量。
- 开发、调试并验证嵌入式控制软件，采用 PLC 编程（Codesys）和工业通信协议（Modbus、CAN、以太网/IP），以确保系统运行的可靠性。
- 设计了新型自动化设备硬件-软件接口的电气原理图和铅成型，确保完全符合行业安全和监管标准（如 IEC 61131）。
- 在一个跨职能研发团队中合作，采用敏捷（Scrum）方法;管理 JIRA 中的项目任务，并用 Git 维护版本控制，以简化开发生命周期。

专业与技术技能

- **AI/ML 与深度学习**
  PyTorch • TensorFlow • Keras • Transformer 架构 • 自监督与多模态学习 • 音视频融合 • 生成模型（扩散模型） • NVFP4 低精度训练 • vLLM • TensorRT-LLM • Transformer Engine • 量化感知训练（QAT） • 分块缩放策略
- **系统与高性能计算**
  CUDA • Triton（内核开发） • NVIDIA Nsight Systems • NVIDIA Nsight Compute • 混合精度训练 • 分布式模型融合 • FP8/FP4 优化
- **编程与数据工程**
  Python（熟练） • C++ • Java • SQL • MATLAB • Apache Spark • Hadoop • Hive • RESTful API
- **云、MLOps 与嵌入式系统**
  AWS • 谷歌云平台（GCP） • Docker • Git • 嵌入式系统（PLC、Codesys、Modbus/CAN/Ethernet/IP）
- **方法与核心能力**
  Agile/Scrum 项目管理 • 跨部门协作 • 完整软件/研究生命周期 • 技术研究与写作 • 复杂问题解决 • 电气原型开发与安全合规 • 性能分析与调试

项目

1. 汽车酒精检测系统 | *嵌入式物联网项目*

- 设计了一款用于汽车安全的实时酒精检测原型，使用红外传感器和 Arduino/ESP32 微控制器。
- 开发了传感器数据处理固件和基于阈值的警报系统，以在检测到时触发警告。
- 设计了硬件接口和外壳，确保车载环境下的可靠性。

2. Apache Spark 分布式机密查询协议 | 保护*隐私的大数据*

- 设计并实现了一个安全、保护隐私的 Apache Spark 查询执行引擎，采用同态加密。
- 在 10 节点集群测试期间，保持 95%的数据机密性，性能开销极低。
- 与基线隐私框架相比，分布式查询性能优化了 20%。

3. 多模态文本转视频生成工具 | *生成式人工智能研究*

- 利用扩散变换器（DiT）架构和 GAN 组件构建了端到端的文本到视频生成流水线。
- 集成跨模态（视听）对齐技术，生成的输出在定性评估中达到 90%的用户满意度。
- 在 AWS EC2 和 S3 上部署了可扩展推理服务，支持按需视频合成。

4. 基于 GAN 的医学图像增强 | *医疗人工智能应用*

- 开发了条件性 GAN 模型，用于超解析和去噪低分辨率 MRI 扫描，提升临床细节。
- 经过标准分割模型验证后，增强后的图像将下游肿瘤检测准确率提升至 85%。

5. 无人机实时视觉物体跟踪 | *计算机视觉与自主系统*

- 设计并训练了一套基于深度学习的无人机视觉跟踪系统，适用于动态环境中的无人机（UAV）。
- 在 BAIR 机器人推力数据集上评估时，实现了 95%的跟踪准确率（IoU），展现出对遮挡的鲁棒性。

认证

- **AWS 认证机器学习——专业 |亚马逊网络服务**
  （验证在主要云平台上发布生产型机器学习解决方案的能力）
- **深度学习专业 | DeepLearning.AI（Coursera）**
  （认证核心 AI/机器学习知识，用于有根据的技术决策和路线规划）
- **敏捷与产品方法论**
  专业 Scrum Master I（PSM I） | Scrum.org
  （展示领导技术团队和管理开发周期的正式敏捷/Scrum 专业知识）

奖项与成就

奖学金与学术荣誉

- 中国政府奖学金（2024 – 2026 |以表彰学业卓越）
- NWPU 总统奖学金（2020 – 2023 |连续颁发，表彰顶尖学业表现）
- 三秦奖学金（2023 年 | 基于优秀科研潜力的奖学金）
- 吴亚军奖学金（2024 年 |基于优秀研究潜力的优异奖）

领导与服务

- 国际学生联盟主任（2023 |领导各项倡议，代表多元化的学生群体）
- 学术志愿者（2022 – 2023 |协助辅导和支持低年级学生的研究）

竞赛与荣誉

- 黑客松获奖者，NWPU AI 挑战赛（2022 年 |带领团队赢得了 48 小时应用人工智能竞赛的胜利）
- 最佳本科生论文奖——*Spark 分布式机密查询协议*（表彰其卓越研究贡献和实施）